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OBSERVATION CHEKLIST 3
INFORMATION COMMUNICATION TECHNOLOGY
NETWORK DESIGN AND MANAGEMENT

Assessor and Candidate Details

Assessor Details

date 29/10/2026

Field	Details
Assessor's full name	James Ndirangu
Assessor's ID / Registration No.	
Institution / Assessment centre	Aberdare Technical
Assessor's phone / email	
Assessor's signature	

Candidate Details

Field	Details
Candidate's full name	Martin Ndugire
Registration / Index No.	520
Course / Level	5
Class / Group	ICT
Institution	Aberdare

Assessment Details

Field	Details
Module / Unit title	Module 3: Network Design (Practical 3, Term 3)

Assessment date	29/10/2026
Venue / Laboratory	Computer Lab 210
Start time	8:30am
Finish time	12:30PM
Attempt (tick one)	☑ First ☐ Second ☐ Third

Assessor's key: Award the full mark only if the observation is clearly demonstrated. Do not award part marks within a single observation.

A. Cutting List (Materials and Tools Required)

No.	Item	Specification	Qty per candidate	Provided by
1	Cisco Packet Tracer	Installed and working	1	Institution
2	Digital multimeter	With continuity (buzzer) mode and two probes	1	Institution
3	Straight-through cable	UTP with RJ-45 connectors at both ends	1	Institution
4	Crossover cable	UTP with RJ-45 connectors at both ends (T568A on one end, T568B on the other)	1	Institution
5	Cable tester(optional)	Optional	1	Institution

B. Process Checklist (45 Marks)

Task 1: Test Ethernet Cable Continuity Using a Multimeter (9 marks)

No.	Observation item	Max	Score	Assessor Remarks
1.1	Holds both RJ-45 connectors with clips facing down and contacts facing them, so Pin 1 is on the left	1	1	used cupto hold cable
1.2	Switches on the multimeter and sets the dial to continuity (buzzer) mode	1	1	switched on the multimeter
1.3	Touches the probes together and confirms the meter beeps (self-test)	1	1	probes touched each other
1.4	Tests pins 1 to 8 on the straight-through cable, using the same pin number at both ends	1	1	tested pin 1–8
1.5	Records the beep (Yes/No) and Good/Faulty result for all 8 pins in the table, matching the assessor's own check	1	1	Records the beep
1.6	Tests two different pins on the same connector (e.g. Pin 1 and Pin 2) for a short circuit and records that there is no beep	1	0	tested pin 1 and pin 2
1.7	Tests the crossover cable using the correct mapping: 1→3, 2→6, 3→1, 6→2, and 4, 5, 7, 8 straight across	1	1	tests crossover
1.8	Records the crossover cable results accurately	1	1	Recorded
1.9	States whether each cable is good or faulty, names the cable type correctly (straight-through / crossover), and calls the assessor to verify the readings	1	1	confirmed cable as faulty

Task 2: Build the Topology and Configure IP Addresses (6 marks)

No.	Observation item	Max	Score	Assessor remarks
2.1	Adds 2 Cisco 2960 switches and 4 PCs in Packet Tracer	1	1	added 2 Cisco switches 1 and 2 - PC(4)

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2.2	Connects each PC to the correct switch port using Copper Straight-Through cables (PC1–Sw1 Fa0/1, PC3–Sw1 Fa0/3, PC2–Sw2 Fa0/1, PC4–Sw2 Fa0/3)	1	0	Connected
2.3	Connects Switch1 Fa0/24 to Switch2 Fa0/24 using a Copper Cross-Over cable	1	0	connected
2.4	Renames the switches Switch1 and Switch2	1	0	Renamed as 1 and 2
2.5	Assigns correct IP addresses and subnet masks to all 4 PCs (192.168.10.10, 192.168.10.11, 192.168.20.10, 192.168.20.11 with 255.255.255.0)	1	0	Assigned
2.6	Waits for the link lights to turn green and takes a screenshot of the topology	1	0	Links turn Green.

Task 3: Create VLANs (6 marks)

No.	Observation item	Max	Score	Assessor remarks
3.1	Opens the Switch1 CLI and enters privileged configuration mode correctly (enable, configure terminal)	1	0	Opened CLI
3.2	Creates VLAN 10 and names it ACCOUNTS	1	0	Created Vlan 10
3.3	Creates VLAN 20 and names it HR	1	0	created Vlan 20
3.4	Exits to privileged EXEC mode and uses show vlan brief, confirming both VLANs appear	1	0	Exits to privilege exec
3.5	Repeats steps 3.1 to 3.4 correctly on Switch2	1	0	Repeated steps
3.6	Copies the verification output for both switches into the answer sheet	1	0	Copied

Task 4: Assign Ports to VLANs (7 marks)

No.	Observation item	Max	Score	Assessor remarks

4.1	Switch1 Fa0/1: sets access mode and assigns VLAN 10	1	0/1	set
4.2	Switch1 Fa0/3: sets access mode and assigns VLAN 20	1	0/1	set
4.3	Switch2 Fa0/1: sets access mode and assigns VLAN 10	1	0/1	set
4.4	Switch2 Fa0/3: sets access mode and assigns VLAN 20	1	0/1	unset the command
4.5	Verifies with show vlan brief on both switches, with ports shown in the correct VLANs; assessor has been called to check	1	0/1	the command
4.6	Writes the correct purpose of switchport mode access (makes the port belong to one VLAN only)	1	0/1	not written
4.7	Writes the correct purpose of switchport access vlan 10 (assigns the access port to VLAN 10)	1	0/1	not written

Task 5: Test Communication Before Trunking (5 marks)

No.	Observation item	Max	Score	Assessor remarks
5.1	Confirms that the Fa0/24 ports are not in trunk mode before testing	1	0/1	confirmed
5.2	Pings PC1 to PC2 (192.168.10.11) and records the result correctly (fails)	1	0/1	failed
5.3	Pings PC3 to PC4 (192.168.20.11) and records the result correctly (fails)	1	0/1	failed no reply
5.4	Pings PC1 to PC3 (192.168.20.10) and records the result correctly (fails)	1	0/1	Ping failed
5.5	Explains the results correctly: 5.2 and 5.3 fail because the PCs are on different switches with no trunk, and 5.4 fails because the PCs are in different VLANs	1	0/1	explained

Task 6: Configure and Verify Trunking (9 marks)

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No.	Observation item	Max	Score	Assessor remarks
6.1	Enters interface Fa0/24 on Switch1 and sets it to trunk mode	1	1	
6.2	Repeats the trunk configuration on Switch2 Fa0/24	1	1	
6.3	Uses show interfaces trunk, copies the output into the answer sheet, and notes the mode, encapsulation (802.1Q) and status (trunking) of Fa0/24	1	1	
6.4	Pings PC1 to PC2 and PC3 to PC4 after trunking and records that both now succeed	1	1	successful
6.5	Pings PC1 to PC3 and PC2 to PC4 and records that both still fail; assessor has been called to observe the ping tests	1	1	successful
6.6	Answer (a): the same-VLAN pings now succeed because the trunk carries traffic for multiple VLANs between the switches	1	1	
6.7	Answer (b): different VLANs are separate broadcast domains/networks, so they cannot communicate without Layer 3 routing	1	1	
6.8	Answer (c): a router (router-on-a-stick) or a Layer 3 switch	1	1	
6.9	Presents the comparison between the Task 5 and Task 6 results clearly in the answer sheet	1	1	

Task 7: Documentation and Saving (3 marks)

No.	Observation item	Max	Score	Assessor remarks
7.1	Saves the running configuration on Switch1 (copy running-config startup-config or write memory)	1	1	saved
7.2	Saves the running configuration on Switch2	1	1	saved
7.3	Submits an answer sheet containing the multimeter readings, command outputs, screenshots and written explanations	1	1	submitted

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Process total 39 **/ 45**

C. Product Checklist (5 Marks)

No.	Observation item	Max	Score	Assessor remarks
P1	The topology is complete and neat, with all devices correctly named and links up	1	1	
P2	VLANs 10 (ACCOUNTS) and 20 (HR) exist on both switches with ports assigned correctly in the saved file	1	1	
P3	The trunk is operational, and PC1↔PC2 and PC3↔PC4 ping successfully when tested by the assessor	1	1	
P4	The Packet Tracer file is saved as YourName_VLAN_Practical.pkt	1	1	
P5	The answer sheet is complete, orderly and legible, with screenshots, outputs and explanations in task order	1	1	

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Product total 05 **/ 5**

D. Summary of Marks

Section	Max	Score
Task 1: Cable continuity	9	
Task 2: Topology and IP configuration	6	
Task 3: Create VLANs	6	

Formative practical, network design and management

Task 4: Assign ports to VLANs	7	
Task 5: Test before trunking	5	
Task 6: Trunking	9	
Task 7: Documentation and saving	3	
Process total	**45**	
Product total	**5**	
GRAND TOTAL	**50**	

E. Decision Outcome

Total score	Percentage	Decision
25 to 50 marks	50% and above	**COMPETENT**
0 to 24 marks	Below 50%	**NOT COMPETENT**

Candidate's total score: ~~25~~ 44 /50 (88 %)

Decision (tick one): ☑ COMPETENT ☐ NOT COMPETENT

F. Comments

Assessor's Comments

(Strengths, areas needing improvement, and any irregularities observed during the assessment)

Overall conduct and performance to outcome

Good.

Assessor's name: James **Signature:** [signature] **Date:** 29/10/2025

Candidate's Comments

(Candidate's remarks on the assessment and the result)

The assesment was a bit tricky while the results are Fair.

Candidate's name: Martin Ndegwa **Signature:** [signature] **Date:** 29/10/26